Situated Knowledge in Knowledge Graphs: A Field-Based Representation of the Qoyllur Rit'i Ritual Landscape

November 3, 2026

Abstract

The Qoyllur Rit'i pilgrimage, an Andean ritual inscribed on UNESCO's Representative List of the Intangible Cultural Heritage of Humanity, remains poorly represented in structured digital systems, where relational and situated forms of knowledge are difficult to encode. Existing data are sparse, institutionally biased, and unable to capture the spatial-temporal organization of ritual landscapes. This paper introduces a field-based approach to knowledge graph construction grounded in ethnographic participation during the 2025 pilgrimage. The resulting graph integrates temporal, spatial, and participatory dimensions with explicit provenance, enabling the representation of relationships that are typically absent from institutional metadata. A key contribution is the modeling of dual event-place entities, which capture the co-constitutive nature of action and space in ritual contexts. We evaluate the graph through a retrieval-based framework, treating retrieval as a diagnostic of representational capacity. Results show that retrieval performance correlates with relational density, revealing both strengths in well-modeled domains and systematic gaps in underrepresented areas. These findings position knowledge graphs not merely as technical tools but as epistemic infrastructures, highlighting the importance of field-based data for addressing structural biases in the digital representation of cultural heritage.

1 Introduction

Cultural heritage is increasingly represented through structured digital systems that shape how knowledge is organized and made visible. In these systems, relational and situated forms of knowledge are difficult to encode, particularly in the case of complex ritual practices. The Qoyllur Rit'i pilgrimage is one of the most complex living ritual systems in the Andes, bringing together tens of thousands of participants across a high-altitude landscape structured by named places, ritual roles, and temporally ordered practices (see Fig. 1). Held annually near the Sinakara glacier in the Cusco region of Peru, the pilgrimage unfolds over five days through processions, nocturnal glacier ascents, communal liturgies, and a 24-hour ritual walk that traverses a network of sacred locations. Its significance extends beyond devotion: Qoyllur Rit'i articulates territory, collective memory, specialized ritual roles, and the intergenerational transmission of Andean cosmological

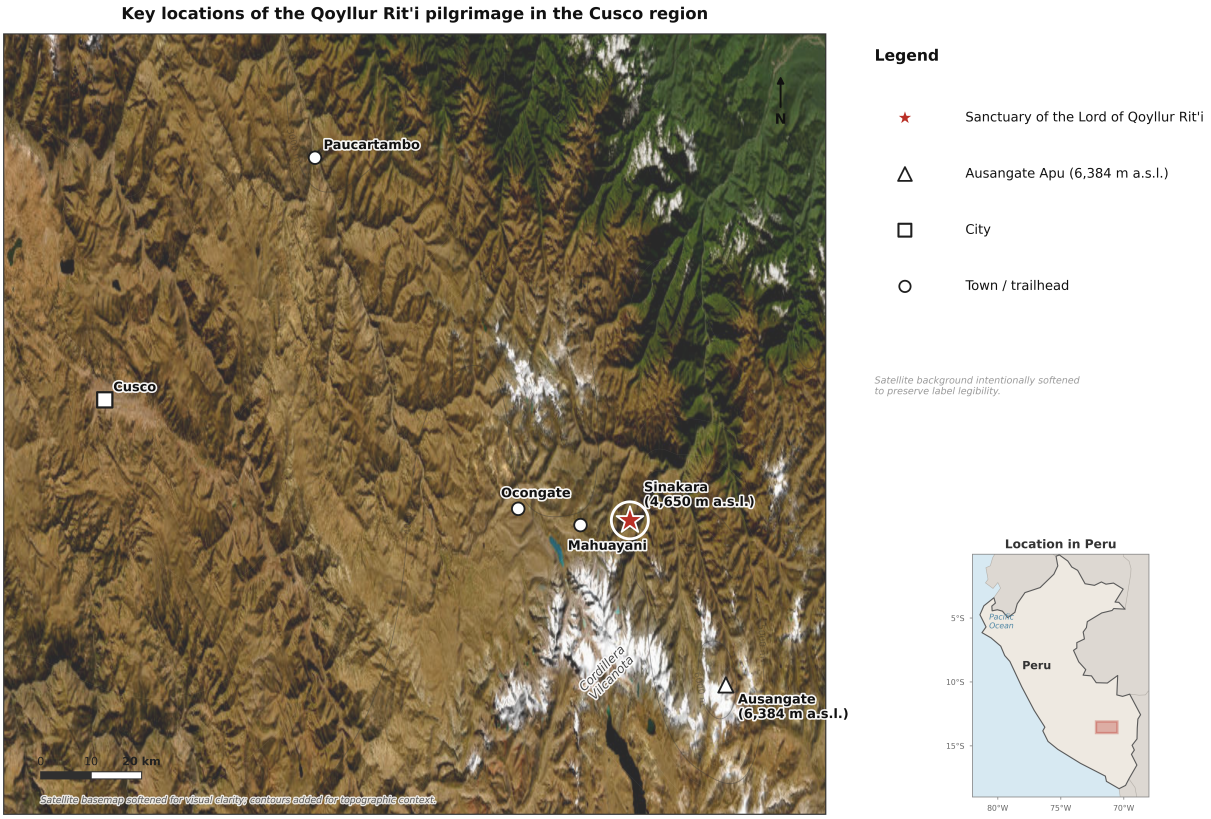

Figure 1: **Key locations of the Qoyllur Rit'i pilgrimage in the Cusco region (Peru).** The figure situates the main spatial references of the pilgrimage within the Vilcanota mountain range, including Cusco, Paucartambo, Mahuayani (the principal trailhead), the Sanctuary of the Lord of Qoyllur Rit'i at Sinakara (4,650 m a.s.l.), and the Ausangate Apu (6,384 m a.s.l.). The satellite basemap is intentionally softened to preserve label legibility, while contour lines derived from interpolated elevation data provide topographic context for the high-altitude ritual landscape. The inset indicates the location of the study area within southern Peru.

knowledge. The retreat of the Sinakara glacier introduces an additional environmental dimension that reshapes ritual practice in real time, situating the pilgrimage at the intersection of indigenous cosmology and climate change, a dimension that remains insufficiently represented in existing digital and institutional catalogues [22, 18]. This paper argues that such limitations are not merely technical but epistemic, and that addressing them requires rethinking how cultural knowledge is produced, structured, and validated in digital systems.

Despite its cultural significance, Qoyllur Rit'i remains poorly represented in structured digital environments, reflecting broader limitations in how heritage data are produced and organized. Two compounding conditions explain this gap. The first is scarcity: web-available information is sparse, institutionally filtered, and semantically flattened, often reduced to brief devotional summaries that obscure the internal diversity of the ritual. Recent network-based analysis of UNESCO intangible heritage metadata for South American countries shows that this scarcity is not incidental but structural: Andean nations are systematically positioned within low-diversity ritual descriptor

clusters, while ecological, cosmological, and epistemic dimensions remain underrepresented [25]. The second condition is dispersion: relevant evidence exists but is fragmented across archives, community records, academic publications, and forms of knowledge transmitted within specific social contexts that rarely enter structured digital systems [2]. Together, these conditions produce a form of informational injustice, in which certain modes of cultural knowledge–particularly those that are embodied, relational, and context-dependent–are systematically excluded from structured digital representation [23, 3, 16].

Knowledge graphs have emerged as a productive infrastructure for addressing this type of representational challenge [14, 1, 7]. By organizing entities, relationships, and evidence within a shared semantic framework, they enable the integration of heterogeneous sources with explicit traceability, moving beyond flat catalogue descriptions toward relational representations of cultural knowledge [5, 29]. Landmark projects such as ArCo [4] and the Chinese intangible cultural heritage multimodal graph [9] demonstrate this potential. However, these efforts share a common epistemic starting point: they are built from institutional repositories and web-available sources. In contexts where such sources are structurally incomplete or biased, this approach risks reproducing rather than correcting existing asymmetries. This limitation motivates an alternative strategy in which knowledge graphs are constructed from the ground up, privileging fieldwork, community knowledge, and situated observation as primary epistemic sources.

Rather than extending existing graph-based approaches built on institutional data, this paper develops an alternative strategy: a field-based knowledge graph constructed from situated, embodied participation. In this approach, fieldwork is not treated as supplementary evidence but as the primary epistemic source. The approach is grounded in direct participation in the Qoyllur Rit'i pilgrimage of June 2025, where the first author took part as a dancer in the ukuku comparsa. The ukuku, also known as ukumari, is a liminal figure that mediates between human and animal worlds, serves as ritual guardian during the glacier ascent, and follows spatial itineraries not described with precision in institutional sources. This participation functions as the primary data collection instrument. The temporal granularity of the ontology, the spatial organization of actors, and the modeling of entities that are simultaneously events and places derive from observations only accessible through embodied participation. Following [6] and recent critical heritage scholarship [11], we argue that this situated position enables ontological distinctions that remain inaccessible to external observation.

This study develops a field-based approach to knowledge graph construction for living heritage systems and evaluates its representational capacity through retrieval-based methods [28, 27]. The remainder of the paper is organized as follows. Section 2 presents the main results, comprising the knowledge graph and its retrieval-based evaluation. Section 3 discusses the findings and their implications for heritage representation and epistemic infrastructure. Section 4 describes the fieldwork design, data acquisition, ontology construction, and ethical protocols.

2 Results

2.1 The Knowledge Graph

2.1.1 Overview and entity statistics

The knowledge graph presented here is not intended as a comprehensive model of the pilgrimage, but as a minimal, field-grounded representation designed to test how relational and situated knowledge can be encoded in structured form.

The Qoyllur Rit'i knowledge graph is a structured semantic representation of the ritual landscape, in which people, places, events, objects, and documentary sources are connected through explicitly typed relationships. Implemented in RDF/Turtle and defined through an OWL ontology, the graph is designed to support machine-readable, interoperable, and extensible representation of field-based heritage knowledge. Rather than storing isolated facts, it encodes the relational organization of the pilgrimage, allowing entities to be connected through temporal sequence, spatial containment, participation, material composition, and provenance. This structure is particularly important in the present case, where the meaning of ritual practices emerges not from individual elements taken separately, but from the patterned relations among actors, places, and events across the pilgrimage.

As shown in Fig. 2, the graph contains 87 named individuals distributed across 15 entity classes, as summarized in Table 1. It includes approximately 340 object property assertions and 180 data property assertions, yielding a total of around 520 triples excluding ontology declarations. The ontology defines 22 object properties, 14 data properties, and 15 classes organized in a shallow hierarchy with no more than two levels of subclass depth. This compact structure reflects a deliberate design choice to maintain ontological clarity while preserving sufficient expressive capacity to capture the relational complexity of the ritual landscape.

Table 1: Distribution of named individuals by class in the knowledge graph.

Class	**English gloss**	**Individuals**
EventoRitual	Ritual event	35
Lugar	Place	39
Vestimenta	Attire	8
Instrumento	Instrument	4
ObjetoRitual	Ritual object	3
Others	—	6
Total		**87**

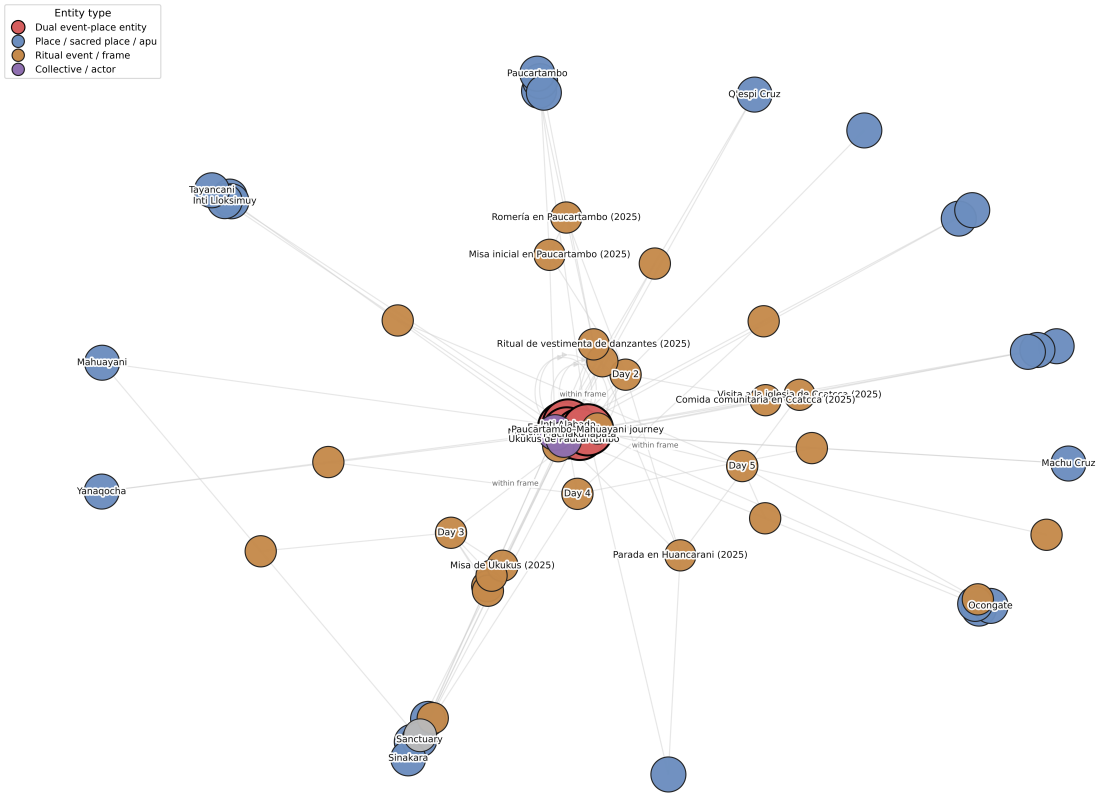

Figure 2: **Relational structure of the Qoyllur Rit'i knowledge graph, centered on the five dual event-place entities.** Red nodes indicate entities modeled simultaneously as *EventoRitual* and *Lugar*, while surrounding nodes represent their immediate and second-order spatial, temporal, and participatory context. The layout is generated using a force-directed algorithm with fixed positions for the dual entities, allowing the global relational structure to emerge while preserving the conceptual centrality of these elements. Edges correspond to selected object property assertions, restricted to the most informative relations for readability.

Two classes overlap in membership: five entities are simultaneously instances of `EventoRitual` (ritual event) and `Lugar` (place), constituting the dual event-place entities introduced in Section 2.1.4 and analyzed in detail in Section 3.2. This overlap is not incidental, but reflects a core ontological feature of the Qoyllur Rit'i ritual landscape, where actions and locations are co-constitutive. In these cases, events cannot be meaningfully separated from the places in which they unfold, and places acquire their significance through the performance of specific ritual actions. Modeling this duality allows the knowledge graph to capture forms of relational meaning that are not representable within conventional categorical distinctions between events and locations.

2.1.2 Temporal structure

The temporal backbone of the knowledge graph organizes the five-day pilgrimage through a two-level hierarchy that reflects both its calendrical structure and its experiential unfolding. At the

upper level, seven named temporal frames structure the progression of the ritual: five diurnal frames corresponding to the named days of the pilgrimage and two nocturnal frames that bridge consecutive days during periods of intensive ritual activity. These frames are modeled as instances of `EventoRitual` (ritual event), with properties such as `tieneOrden` (sequence order) encoding their position within the overall progression and `tieneFecha` (calendar date) anchoring them to specific dates in June 2025.

The five diurnal frames are `Día 1: Sábado de preparación` (June 14), `Día 2: Domingo de partida y viaje` (June 15), `Día 3: Lunes de ascenso al santuario` (June 16), `Día 4: Martes de descenso e inicio de la lomada` (June 17), and `Día 5: Miércoles del alba y finalización` (June 18). The two nocturnal frames, `Noche del lunes al martes` and `Noche del martes al miércoles`, correspond to phases of intensified ritual activity, including the glacier ascent and the central segment of the Lomada. Together, this structure captures a temporal organization in which diurnal and nocturnal cycles are not secondary subdivisions, but integral components of the ritual sequence.

At the lower level, 28 discrete events are linked to their corresponding temporal frames through the `defineMarcoTemporal` property. Each event is associated with a `tieneOrdenEvento` (event order) attribute that encodes its sequential position within the frame, allowing the knowledge graph to represent not only which events occur on a given day, but the order in which they unfold. This structure captures a wide range of activities, from relatively simple actions such as the communal meal at the `Plaza de Ccatcca` to more complex ritual sequences including the `Misa de Ukukus`, the ascent to `Colque Punku`, and the ceremonial entry through `Pachakunapata`. Event duration is encoded where available through the `tieneDuracionHoras` property, with the `Lomada` extending over 24 hours, the glacier ascent lasting approximately 6.5 hours, and the descent from `Colque Punku` to the sanctuary taking around 3 hours. Together, these attributes allow the graph to represent the temporal texture of the pilgrimage at a finer granularity than day-level descriptions alone.

Temporal ordering between events is encoded through two complementary properties. The `antesDe` (before) relation, defined as transitive, captures general precedence relationships among events within the same temporal frame, while `sucedeInmediatamenteA` (immediately follows), defined as non-transitive, encodes direct succession without intermediate events. This distinction allows the graph to differentiate between loose temporal ordering and strict sequential continuity, supporting queries that depend on both types of temporal relations. As a result, the knowledge graph can represent sequence, precedence, and temporal proximity in ways that are not accessible through flat metadata structures.

2.1.3 Spatial structure

The spatial backbone of the knowledge graph organizes 39 named places through a hierarchy of containment relationships encoded via the transitive `estaEn` (is located in) property and its inverse `contiene` (contains). This transitivity is ontologically significant, as it enables the graph to repre-

sent nested spatial relations across multiple scales. For example, if Sinakara is located within the Ausangate and the Santuario del Señor de Qoyllur Rit'i is located within Sinakara, then the santuario is transitively situated within the Ausangate. This structure allows retrieval queries to navigate the full geographic and cosmological context of any place within the ritual landscape.

At the apex of the spatial hierarchy is the Ausangate, modeled as an instance of Apu (mountain spirit), a subclass of LugarSagrado (sacred place). The concept of apu in Andean cosmology refers to entities that are simultaneously physical mountains, sacred presences, and active cosmological agents. The Ausangate, at 6,384 meters above sea level, is the highest peak in the Cusco region and functions as both the geographic and cosmological container of the pilgrimage. It directly contains Sinakara, the sacred mountain area at 4,650 meters where the sanctuary is located, and Colque Punku, the glacier at 5,200 meters that constitutes the destination of the nocturnal ritual ascent.

The Santuario del Señor de Qoyllur Rit'i is itself modeled as a spatial container, comprising three named sub-places: Mamachapata, the processional area toward which the image of the Señor de Tayankani is carried; Pachakunapata, the ceremonial entry threshold; and the Celda de los Ukukus de Paucartambo, a bofedal encampment where the ukuku comparsa rests and prepares prior to the glacier ascent. This internal structure reflects spatial distinctions that emerge from participation in the ritual landscape, rather than from cartographic or institutional sources. To our knowledge, no available map or official documentation captures these zones and their associated ritual functions in comparable relational detail.

Beyond the sanctuary and glacier, the spatial network extends throughout the entire pilgrimage route followed by the Nación Paucartambo. The outbound route begins in Paucartambo, passes through Huancarani, Ccatcca, and Ocongate, reaches the trailhead at Mahuayani, and ascends to the sanctuary. The return route follows a distinct path through Yanaqancha, Yanaqocha, Q'espi Cruz, Qquchiyoc Wayq'o, Escalerachayoq, Inti Lloksimuy, and Tayancani, before concluding at the Plaza de Ocongate. Each of these locations is georeferenced and linked to the events that occur within them through the ocurreEnLugar (occurs in place) property, allowing the graph to represent the pilgrimage as a continuous spatial trajectory rather than a set of isolated sites.

In addition to the Ausangate, eight locations are classified as LugarSagrado (sacred place), including Colque Punku, Sinakara, Machu Cruz, Q'espi Cruz, Yanaqocha, and the sanctuary itself. This classification reflects their cosmological significance within the ritual geography and allows the graph to distinguish between ordinary locations and those endowed with specific ritual and symbolic roles.

2.1.4 Dual event-place entities

The five dual event-place entities constitute the central ontological contribution of this work, emerging directly from embodied fieldwork within the ritual landscape. Each entity is modeled simultaneously as an instance of EventoRitual (ritual event) and Lugar (place), integrating temporal and spatial properties within a single representation. As events, they carry attributes such as tieneDuracionHoras (duration in hours) and tieneOrdenEvento (event order), while as places,

they are defined through spatial relations including `estaEn` (is located in), `conduceA` (leads to), `pasaPor` (passes through), and geolocation. This dual modeling enables the graph to represent entities whose meaning emerges from the inseparability of action and location, rather than forcing an artificial distinction between the two.

The `Lomada`, also known as *Loman Pureq* ("those who walk the hills"), is the most elaborated of these entities. As an event, it is a 24-hour ritual walk performed by `Nación Paucartambo`, beginning at the sanctuary and ending at the Inti Alabado site at dawn. As a place, it is the route itself, composed of a sequence of seven named waypoints encoded through `pasaPor` relations: `Machu Cruz`, `Yanaqocha`, `Yanaqancha`, `Qquchiyoc Wayq'o`, `Q'espi Cruz`, `Escalerachayoq`, and `Inti Lloksimuy`. Directionality is captured through `desde` (from) and `conduceA` (leads to), linking the `SantuarioQoylluriti` to `IntiAlabado`.

This dual representation is not merely analytical. The `Lomada` is experienced as a structured trajectory in which different participants assume distinct spatial and ritual obligations along the route, and where each waypoint constitutes a differentiated site of practice. Modeling the Lomada as both an event and a place allows the knowledge graph to capture this distributed and relational form of meaning.

`Inti Alabado` is simultaneously the solar greeting ceremony performed at dawn on the fifth day and the high-altitude site in `Tayancani`, where participants gather to await sunrise. Its Quechua name, *Inti Lloksimuy*, refers to the emergence of the sun, and the location is traditionally understood as the point at which the ritual sequence culminates in a solar transition. As an event, it is performed by texttttNación Paucartambo and occupies the first position (`tieneOrdenEvento` = 1) within Día 5. As a place, it is geolocated within the Tayancani sector and linked to the broader spatial hierarchy through `estaEn`. This dual encoding reflects the fact that the ceremonial act and the site in which it occurs are inseparable components of a single ritual moment.

`Pachakunapata` is simultaneously the ceremonial entry ritual through which the Nación enters the sanctuary as a unified collective and the named threshold space at the sanctuary entrance where this transition occurs. The ritual involves a period of waiting, preparation, and coordinated movement before formal entry, a sequence that is only fully intelligible when `Pachakunapata` is understood as both action and place. This dual encoding captures the spatially anchored nature of collective transformation within the ritual landscape.

`Escalerachayoq` is both a segment of the Lomada traversed between three and four in the morning and the specific topographic feature that defines that segment, a rocky formation resembling a stone stairway. Its Quechua name, *Escalerachayoq*, means "the one that has stairs." As a waypoint, it is encoded through the `pasaPor` relationship within the Lomada route; as an event, it is modeled as a sub-event within the nocturnal Lomada frame, carrying its own `tieneOrdenEvento` value. This dual representation reflects how particular sections of the route are experienced not merely as locations but as temporally and physically distinct moments within the progression of the pilgrimage.

The vehicular journey from `Paucartambo` to `Mahuayani` constitutes the fifth dual entity. As an

event, it represents the collective departure of `Nación Paucartambo` on the second day of the pilgrimage, structured through ritualized stops at the casa del prioste and the `iglesia de Ccatcca`. As a place, it is the route itself, encoded through `desde` (Paucartambo), `conduceA` (Mahuayani), and `pasaPor` relations linking intermediate locations such as `Huancarani`, `Ccatcca`, and `Ocongate`. Modeling this journey as both event and place reflects a broader Andean spatial logic in which routes between communities are not neutral connectors but meaningful entities with their own ritual and social significance.

2.1.5 Ritual actors and objects

The knowledge graph also encodes relationships among ritual actors and material objects, capturing additional layers of relational structure. For example, `Nación Paucartambo` and its ukuku comparsa are represented as collective entities linked to events through participation relations, while `Traje Ukumari` is modeled as a composite object with multiple constituent parts connected through `tieneParte` relationships. This compositional encoding allows the graph to represent not only ritual actions and spaces, but also the material configurations through which those actions are performed, enabling retrieval queries that traverse from actors to objects and their symbolic components within a unified relational framework.

2.2 Graph-RAG Evaluation

2.2.1 Pipeline architecture

The Graph-RAG pipeline implemented in this work is conceived as an evaluation instrument for the knowledge graph rather than as a standalone retrieval system. It follows a three-phase architecture that combines hybrid retrieval, context construction, and controlled generation [8, 20], as summarized in Figure 3. In the first phase, candidate entities are retrieved through a hybrid strategy that integrates semantic similarity based on sentence embeddings with lexical matching over normalized entity labels and descriptions [20]. Entity representations are constructed from labels, comments, type annotations, and immediate relational neighborhoods, and encoded using the `paraphrase-multilingual-MiniLM-L12-v2` model [21], which produces 384-dimensional multilingual embeddings. At query time, semantic similarity is computed via cosine similarity, while lexical relevance is assessed through term overlap. The two scores are combined through a weighted scheme ($\alpha = 0.6$ for semantic and 0.4 for lexical), with additional boosting rules for exact matches and spatially related entities. The top $k = 10$ candidates are then passed to the context-building phase.

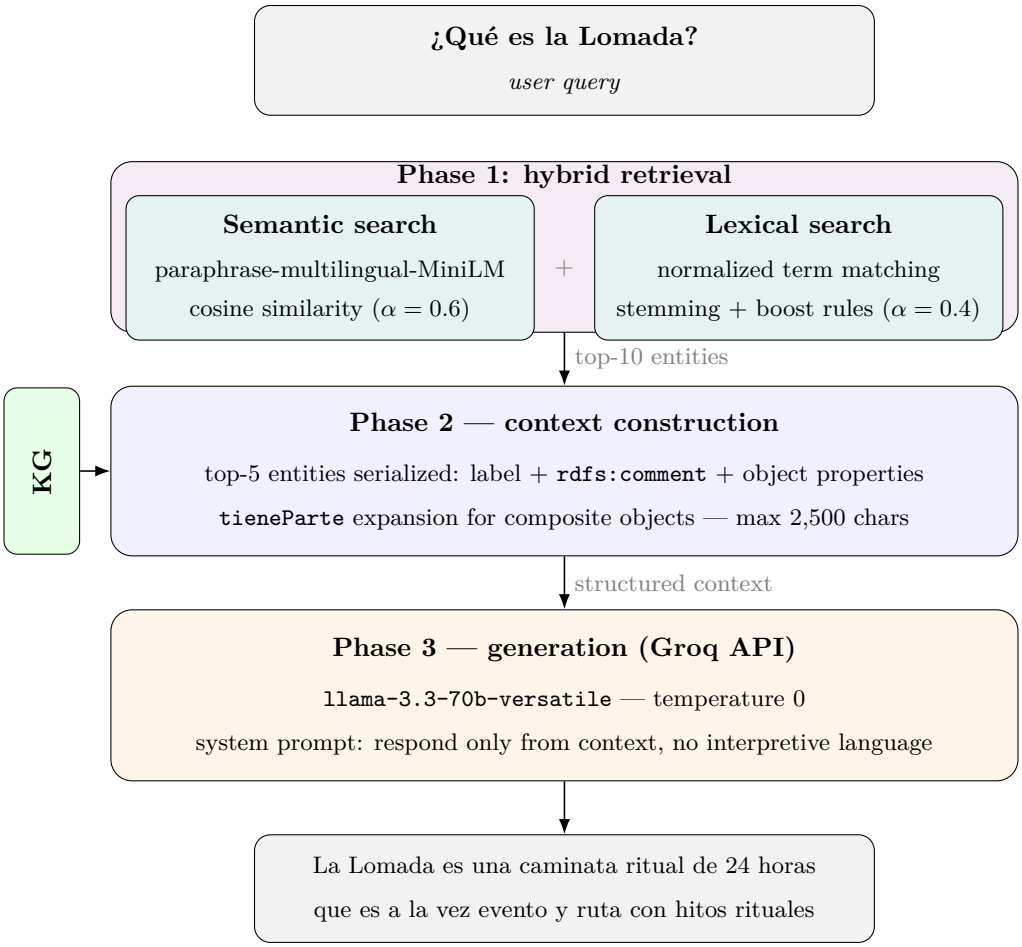

Figure 3: **Architecture of the Graph-RAG pipeline, illustrated with a real query and response from the evaluation set.** The hybrid retrieval phase combines semantic embedding similarity and lexical matching to select the top-k entities from the knowledge graph (KG). The context construction phase serializes the retrieved subgraph into structured natural language. The generation phase uses a large language model conditioned on the constructed context to produce a traceable, grounded response.

In the second phase, the top-ranked entities are serialized into a structured natural-language context designed to preserve relational information while remaining interpretable to the language model. Each entity is represented through its label, a truncated description derived from its `rdfs:comment`, and a selection of relevant object property assertions translated into natural language via a predefined mapping. For composite entities, component structures are expanded through `tieneParte` relations. The resulting context is constrained to a maximum of 2,500 characters to balance informational richness with the effective input limits of the language model.

In the third phase, the constructed context and the original query are submitted to a language model through the Groq API using the `llama-3.3-70b-versatile` model at temperature 0. The system prompt constrains the model to generate responses exclusively grounded in the provided context, prioritizing specific relational information over general definitions and avoiding interpretive

or speculative language not supported by the graph. The pipeline is implemented in Python and deployed as a public Streamlit application (https://qoyllurity.streamlit.app/), with the full codebase available at https://github.com/javier-vz/qoyllurity.

2.2.2 Retrieval performance and limitations

The Graph-RAG evaluation demonstrates that the knowledge graph supports accurate and contextually rich retrieval for queries grounded in explicitly modeled relational structures. Temporal queries are consistently well resolved, and the system correctly identifies event sequences, durations, and transitions between diurnal and nocturnal frames. Spatial queries similarly benefit from the hierarchical organization of places, allowing the system to retrieve multi-scale geographic context, from specific waypoints to broader ritual zones.

Participatory queries show mixed performance. While collective actors such as `Nación Paucartambo` and `ukuku comparsa` are correctly identified and linked to events, the graph's limited representation of individual-level variation constrains the system's ability to answer more granular questions about roles and responsibilities. Material queries related to ritual objects, particularly `Traje Ukumari`, demonstrate strong performance due to the compositional structure encoded through `tieneParte` relationships, enabling the recovery of detailed inventories of object components.

However, the evaluation also reveals clear limitations. Queries that depend on implicit knowledge, unmodeled relationships, or broader contextual interpretation tend to produce incomplete responses. In particular, the system struggles with questions that require linking across weakly connected subgraphs or that rely on knowledge not explicitly encoded in the current version of the graph. These limitations are not artifacts of the retrieval pipeline, but direct reflections of the graph's current scope and density.

2.2.3 Evaluation design

We designed a structured evaluation set of 16 queries organized into four categories, each targeting a distinct retrieval capability of the knowledge graph. The categories were motivated by the structural properties of the graph identified in Section 2.1.

Type A queries test spatial retrieval, asking about the location, identity, and geographic relationships of places in the knowledge graph. These queries are expected to perform well given the dense spatial hierarchy encoded through `estaEn`, `contiene`, and `esParteDelApu` relationships, and the altitude annotations available for key sites.

Type B queries test temporal and sequential retrieval, asking about durations, event sequences, and the order of ritual phases within the pilgrimage. These queries exercise the `defineMarcoTemporal`, `tieneOrdenEvento`, and `antesDe` properties, as well as the named day and nocturnal frame structure described in Section 4.2.

Type C queries target the dual event-place entities, asking questions whose correct answer requires understanding that certain entities are simultaneously actions and places. These queries are

designed to reveal whether the ontological duality encoded in the graph is successfully transmitted through the retrieval and generation pipeline.

Type D queries deliberately target knowledge that is absent or underrepresented in the first-degree graph, including other participating nations, the cosmological significance of the glacier ascent, and the relational role of the `Ausangate` as an active entity rather than a geographic container. These queries function as a diagnostic of the graph's current coverage limits and as a direct motivation for the fieldwork planned for 2026.

The 16 queries are listed in Table 2.

Table 2: Evaluation query set organized by category.

Type	Q	Query (original)	English translation
A	1	¿Dónde ocurre la festividad del Señor de Qoyllur Rit'i?	Where does the festival of the Lord of Qoyllur Rit'i take place?
A	2	¿Dónde está Sinakara?	Where is Sinakara?
A	3	¿Dónde está el glaciar Colque Punku?	Where is the Colque Punku glacier?
A	4	¿Qué es el Ausangate?	What is the Ausangate?
B	5	¿Cuánto dura la celebración?	How long does the celebration last?
B	6	¿Qué eventos ocurren el lunes?	What events take place on Monday?
B	7	¿Qué lugares recorre la Lomada?	What places does the Lomada traverse?
B	8	¿Qué pasa antes de llegar al santuario?	What happens before arriving at the sanctuary?
C	9	¿Qué es Qoyllur Rit'i? Describe la festividad.	What is Qoyllur Rit'i? Describe the festival.
C	10	¿Qué ocurre en el Inti Alabado?	What happens at the Inti Alabado?
C	11	¿Qué es la Lomada?	What is the Lomada?
C	12	¿Dónde ocurre Pachakunapata?	Where does Pachakunapata take place?
D	13	¿Qué hacen los ukumari?	What do the ukumari do?
D	14	¿Qué otras naciones participan en Qoyllur Rit'i?	What other nations participate in Qoyllur Rit'i?
D	15	¿Por qué suben los ukumari al glaciar?	Why do the ukumari ascend the glacier?
D	16	¿Qué relación tiene el Ausangate con la festividad?	What is the relationship between the Ausangate and the festival?

2.2.4 Retrieval-based evaluation

Table 3 summarizes the retrieval and generation outcomes for all 16 queries. For each query we report the top-1 retrieved entity with its hybrid score, whether the top-5 retrieval set contained ontology metadata nodes (TBox noise) rather than named individuals (ABox entities), and a qualitative assessment of the generated response.

Table 3: Summary of retrieval and generation results across the 16-query evaluation set. TBox noise indicates that ontology metadata nodes (classes or properties) appeared among the top-5 retrieved entities. Response quality is assessed as correct (C), partial (P), or failed (F).

Q	Top-1 entity	English gloss	Score	TBox	Resp.
A1	`Festividad del Señor de Qoyllur Rit'i`	Festival of the Lord of Qoyllur Rit'i	2.889	No	C
A2	`Sinakara`	Sinakara	1.614	Yes	C
A3	`Glaciar Colque Punku`	Colque Punku glacier	2.877	No	C
A4	`Ausangate`	Ausangate	0.791	No	C
B5	`tiene duración en horas` (property)	has duration in hours (property)	1.448	Yes	C
B6	`documenta evento` (property)	documents event (property)	1.429	Yes	F
B7	`Lomada / Loman Pureq (2025)`	Lomada / Loman Pureq (2025)	1.525	Yes	P
B8	`Santuario del Señor de Qoyllur Rit'i`	Sanctuary of the Lord of Qoyllur Rit'i	1.414	Yes	C
C9	`Festividad del Señor de Qoyllur Rit'i`	Festival of the Lord of Qoyllur Rit'i	2.370	No	C
C10	`Inti Alabado (2025)`	Inti Alabado (2025)	2.233	No	C
C11	`Lomada / Loman Pureq (2025)`	Lomada / Loman Pureq (2025)	1.518	No	C
C12	`P'achakunapata`	P'achakunapata	0.700	No	P
D13	`Ukumari (Ukuku)`	Ukumari (Ukuku)	1.496	No	C
D14	`Festividad del Señor de Qoyllur Rit'i`	Festival of the Lord of Qoyllur Rit'i	1.456	Yes	F
D15	`Noche del lunes al martes`	Monday-to-Tuesday nocturnal frame	1.368	No	P
D16	`tiene fecha` (property)	has date (property)	1.891	Yes	F

Type A queries produced correct responses in all four cases. The pipeline successfully navigated the spatial hierarchy of the knowledge graph, correctly identifying that the festival occurs in Sinakara within the Ausangate, that Colque Punku is a glacier within the same apu, and that the Ausangate is a mountain spirit at 6,384 meters. Notable in A4 is that the top-1 score for the Ausangate query was only 0.791, the lowest in the entire evaluation set, and yet the response was correct because the Ausangate entity carried sufficient relational context in its neighborhood to ground an adequate answer.

Type B queries produced mixed results. Query B5 (duration) succeeded despite TBox noise in

the top-5 set: the property node *tiene duración en horas* appeared as top-1, but the Festividad entity with its `tieneDuracionHoras` value of 120 hours appeared fourth in the ranking and provided the grounding for a correct response. Query B6 (events on Monday) failed completely. The top-5 retrieval set was dominated by ontology metadata nodes (*documenta evento*, *EventoRitual*, *define marco temporal*) with no named individual among the top results. The node *Día 3: Lunes de ascenso al santuario*, which contains the relevant `defineMarcoTemporal` relationships to six ordered events, was not retrieved because the word *lunes* does not appear as a high-weight term in the entity text constructed for that node, whose label foregrounds *ascenso al santuario* rather than the day name. Query B7 (Lomada route) was partially successful: the correct entity was top-1 and three of the seven `pasaPor` waypoints appeared in the response, but Yanaqancha and Escalerachayoq were omitted because the context construction phase truncated the relational neighborhood at three object properties per entity. Query B8 (before the sanctuary) correctly identified the Caminata Mahuayani-Santuario and the Día 3 frame.

Type C queries demonstrated that dual event-place entities are recoverable through the pipeline, but with variable fidelity depending on relational density. Query C11 (what is the Lomada) produced the best response in the entire evaluation set: the system explicitly verbalized the event-place duality, describing the Lomada as simultaneously a ritual action and a physical route with named waypoints. This result is directly attributable to the explicit duality statement in the `rdfs:comment` of the Lomada entity and the seven `pasaPor` relationships that constitute its spatial identity. Query C10 (Inti Alabado) captured the duality implicitly, describing the solar greeting ceremony performed at the homonymous site. Query C12 (Pachakunapata) produced a circular response: the system answered that Pachakunapata occurs at the Santuario, which contains Pachakunapata, correctly identifying the containment relationship but failing to articulate the event-place duality. The contrast between C11 and C12 is instructive: both entities have duality encoded in their `rdfs:comment`, but the Lomada has seven spatial relationships while Pachakunapata has one, and this relational density difference is directly reflected in response quality.

Type D queries revealed the first-degree graph's coverage limits as expected. Query D13 (what do ukumari do) was the most informationally rich response in the evaluation: the pipeline retrieved the Ukumari class, the DanzaUkumari, the TrajeUkumari with its nine component parts, and the Ukumaris de Paucartambo individual with its participaEn link to the Lomada, producing a response that accurately described role, dress, and participation. Query D14 (other nations) failed cleanly because only one nación is modeled in the current graph. Query D15 (why do ukumari ascend the glacier) produced a partial response: the system correctly identified that the ascent occurs during the nocturnal frame dedicated to glacier rituals but could not provide the cosmological motivation for the ascent because no triples encode the relationship between the glacier, the Ausangate as active cosmological agent, and the ritual obligations of the ukuku role. Query D16 (Ausangate relation to festival) produced the most technically revealing failure in the evaluation: the top-1 retrieved entity was the OWL data property *tiene fecha* with a score of 1.891, the highest score in the entire Type D set. The Ausangate entity, which is linked to the festival through the transitively

reachable `ocurreEnLugar` chain, was not retrieved because the relational query framing activated property-level semantics in the embedding space rather than individual-level semantics.

3 Discussion

The central claim of this study is that representation in knowledge graphs is not a neutral encoding process, but a form of epistemic selection that determines what aspects of cultural systems become computationally accessible.

3.1 Retrieval as diagnostic of graph structure

Interpreting retrieval as a diagnostic process allows the evaluation to move beyond performance metrics and toward an analysis of representational capacity. The results indicate that the knowledge graph effectively captures relational structures that are explicitly grounded in fieldwork, particularly those involving temporal sequences, spatial hierarchies, and dual event-place entities. These dimensions exhibit high relational density and yield coherent retrieval contexts.

Conversely, areas of weaker performance correspond to zones of lower graph density or incomplete modeling, particularly in the representation of actor differentiation and broader cultural context. These gaps highlight not only limitations of the current graph, but also priorities for future fieldwork and data integration.

From this perspective, Graph-RAG functions as a tool for reflexive analysis of the knowledge graph itself. Rather than treating retrieval failures as system errors, they can be understood as indicators of epistemic absence or under-representation. This reframing aligns with the broader aim of the present work: to construct knowledge graphs that do not merely reproduce available data, but make visible both what is known and what remains to be documented within the ritual landscape.

Three technical findings emerge from the evaluation results with implications for the design of future knowledge graph versions and retrieval pipelines in heritage contexts.

TBox/ABox contamination. The entity index used for retrieval includes both ontology metadata nodes (classes, properties, and annotation axioms from the TBox) and named individuals (the ABox). In seven of the sixteen queries, TBox nodes appeared among the top-5 retrieved entities, and in three cases (B6, D16, and partially B5) they displaced relevant individuals from the retrieval set. This contamination is particularly acute for relational queries, where the semantic embedding of a property node (for example
texttttiene fecha or `define marco temporal`) closely matches the embedding of a query asking about relationships. The straightforward mitigation is to filter TBox nodes from the retrieval index, restricting search to named individuals only.

Relational density and dual-entity fidelity. The contrast between the Lomada (C11, correct with explicit duality verbalization) and Pachakunapata (C12, circular response without duality) demonstrates that response quality for dual event-place entities is determined not by

their ontological type but by the density of their relational neighborhood. The Lomada has seven `pasaPor` relationships, one `desde`, one `conduceA`, and an explicit duality statement in its comment; Pachakunapata has one `estaEn` and one `ocurreEnLugar` relationship pointing to the same entity. This finding has a direct implication for the 2026 fieldwork agenda: densifying the relational neighborhood of underspecified dual entities is as important as adding new entities.

Semantic entry-point dependency. Query A1 correctly identifies the Ausangate as a location of the festival by retrieving the Festividad entity and traversing its `ocurreEnLugar` relationships. Query D16 asks directly about the Ausangate's relation to the festival and fails because the retrieval entry point shifts from the festival to the relationship itself, activating TBox semantics. The same information is present in the graph in both cases, but the retrieval outcome depends on which entity serves as the semantic anchor. This asymmetry suggests that for heritage knowledge graphs with small to medium entity counts, adding direct inverse relationships between central entities (for example, an explicit `esContextoCosmologicoDe` from the Ausangate to the Festividad) would make the graph more robust to query formulation variation.

3.2 Ontological modeling and dual entities

The results of this work position knowledge graph construction not merely as a technical task, but as an epistemological practice. By grounding the graph in field-based, situated knowledge, the present approach departs from dominant paradigms in which heritage data are derived primarily from institutional repositories and web-available sources. This shift has direct consequences for what becomes representable: rather than reproducing existing metadata structures, the graph encodes relationships that emerge from participation in the ritual landscape itself.

The modeling of dual event-place entities illustrates this point most clearly. Conventional ontologies enforce a separation between events and locations, reflecting a categorical distinction that does not hold in many Andean ritual contexts. By allowing entities to be simultaneously instantiated as both, the graph captures forms of meaning that are otherwise lost in translation to standard metadata schemas. This suggests that ontological flexibility is not a technical convenience, but a requirement for representing living cultural systems whose structure is relational, processual, and context-dependent.

The Graph-RAG evaluation further demonstrates that retrieval performance is inseparable from representational design. Rather than treating retrieval as a downstream task, the evaluation shows that what can be retrieved is determined by what has been modeled. In this sense, retrieval functions as a diagnostic lens on the knowledge graph, revealing both its strengths and its blind spots. Areas of high performance correspond to regions of dense relational encoding, while incomplete responses point to zones of underrepresentation that require further fieldwork and data integration.

A further implication of the evaluation concerns not only graph density, but ontological scope. Some of the weakest responses are associated with queries that require access to ritual motivations and distributed agency rather than to event structure alone. In particular, the partial response to the glacier-ascent query and the failure to articulate the relation between the `Ausangate` and

the festival suggest that the current graph still encodes where and when ritual actions occur more effectively than why they are performed or how agency is distributed across human and non-human participants. This limitation points to an ontological gap rather than a purely retrieval-level failure. Related work in cultural heritage and digital humanities has emphasized the importance of modelling interpretive and contextual dimensions beyond structural description [6, 13]. Addressing this gap would likely improve not only retrieval performance for cosmological and motivational queries, but also the graph's capacity to represent the internal logic of the ritual landscape.

3.3 Knowledge graphs as epistemic infrastructures

These findings suggest that knowledge graphs should be understood not only as data structures but as epistemic infrastructures that actively shape cultural visibility in AI-mediated environments. If knowledge graphs are increasingly used as substrates for AI systems, then their construction cannot be understood as neutral. The choices made at the level of ontology, data collection, and provenance encoding directly influence how cultural knowledge is retrieved, interpreted, and circulated. A field-based approach, as proposed here, offers one pathway toward mitigating representational asymmetries by incorporating community knowledge and embodied experience into the structure of the graph itself.

3.4 Limitations and future directions

At the same time, this approach raises important challenges. Field-based knowledge graph construction is resource-intensive, context-specific, and difficult to scale compared to automated extraction from large corpora, as it relies on sustained engagement with communities and careful ethical protocols. These conditions constrain its scalability: it cannot be extended without risking the loss of the epistemic specificity that constitutes its main contribution. Future work should therefore explore how such approaches can be combined with other data sources and adapted across contexts while preserving the situated knowledge that gives them their value.

This paper presented a field-based approach to knowledge graph construction for living ritual systems, using the Qoyllur Rit'i pilgrimage as a case study. By grounding the graph in situated knowledge derived from participation, interviews, and direct observation, the resulting representation captures relational structures that are absent from institutional metadata and web-based sources.

The introduction of dual event-place entities constitutes the main ontological contribution of this work, demonstrating the need for representational models that accommodate the co-constitutive nature of action and space in ritual contexts. The Graph-RAG evaluation further showed that retrieval performance reflects the underlying structure of the graph, highlighting both its representational strengths and its current limitations.

More broadly, the results suggest that knowledge graphs for cultural heritage should be understood as epistemic infrastructures rather than neutral data structures. Their design shapes not only what information is stored, but what knowledge becomes visible and retrievable in AI-mediated

environments. Incorporating field-based and community-derived knowledge into these systems is therefore not only a methodological choice, but an ethical imperative.

Future work will extend the knowledge graph through additional field campaigns and comparative analysis with other Andean ritual systems, with the aim of refining both the ontology and the evaluation framework. By making both the graph and its construction process explicit, and by drawing on emerging data-driven approaches to cultural heritage annotation and structuring [17], this work contributes to the development of more transparent, reflexive, and context-sensitive approaches to digital heritage representation.

A second line of future work concerns the explicit modelling of ritual motivations and distributed agency within the knowledge graph. The current version captures events, places, actors, and their relations, but it does not yet represent purposes such as penance, intercession, or collective service as independent semantic nodes. Extending the ontology with additional structures linking events to their attributed purposes and beneficiaries would allow the system to encode why specific ritual actions are performed and on what evidentiary basis those interpretations are made. Such extensions are aligned with recent work on knowledge graphs for cultural heritage, which emphasizes the integration of richer semantic layers and interpretive context [14, 10]. More broadly, this would allow the graph to move beyond structural description toward a more expressive representation of ritual motivations and human/non-human participation in the pilgrimage.

An additional line of future work concerns the adaptation of the embedding model to the domain-specific structure of the knowledge graph. As noted in related work, dense retrieval systems rely critically on the quality of the embedding space used to represent queries and entities [21, 15]. In the present case, multilingual models trained on general web corpora underrepresent Quechua and domain-specific ritual terminology, which may limit retrieval performance. A promising direction is to leverage the relational structure of the knowledge graph itself as a source of weak supervision for contrastive fine-tuning, using connected entities as positive pairs and structurally distant nodes as negatives. This approach would allow the embedding model to better capture the semantic organization of the ritual domain, potentially improving both retrieval accuracy and the interpretability of Graph-RAG responses [12, 8].

4 Methods

4.1 Fieldwork design

This study is based on fieldwork conducted during the Qoyllur Rit'i pilgrimage in June 2025, complemented by prior research on the Virgen del Carmen festival in Paucartambo. This prior engagement provided the community relationships and domain familiarity necessary for data collection in the Qoyllur Rit'i context.

Field access was established through collaboration with a ukuku comparsa from Paucartambo, composed of participants directly embedded in the ritual organization of the pilgrimage. The first author participated in the pilgrimage as a ukuku dancer, enabling direct observation of spatial

itineraries, temporal sequences, and role-specific practices that are not documented in institutional sources. A co-author based in Cusco contributed to the design of data collection and to the interpretation of locally grounded knowledge. In addition, one co-author had participated in the ukukus in his youth and provided close guidance throughout the fieldwork and interpretation process.

All data collection followed informed consent protocols. Participants were informed of the academic purpose of the project, the non-commercial use of the data, and their right to withdraw at any time. Personal identifiers were recorded only with explicit permission. These procedures follow established ethical guidelines for cultural heritage research and AI-related data practices [23].

4.2 Data acquisition

Data acquisition followed a model-first, field-synchronous strategy in which observations were encoded into ontological structures during or immediately after fieldwork. Rather than storing field notes as unstructured text, each observation was recorded through structured annotations linked to predefined entity types and relations. At minimum, each observation was associated with an event identifier, an actor or collective, a named place, and metadata specifying evidence type and date.

Data entry was performed through structured tagging on mobile devices during the pilgrimage, followed by formal consolidation into the ontology after each fieldwork session. This procedure was designed to maintain consistency between field observations and the evolving graph structure.

Photographic documentation was treated as structured data rather than supplementary material. Each image was annotated with references to the event, place, and participants depicted, and modeled as an instance of a multimedia entity linked to the corresponding nodes in the graph.

Interviews with ritual participants and authorities were segmented into discrete statements, each mapped to entities and relations in the ontology. Each statement retained a reference to its source interview, enabling traceability of all assertions.

Documentary sources, including the UNESCO inscription [24] and related institutional reports, were used as contextual references during data interpretation. These sources were not directly encoded as structured entities in the knowledge graph, but were used to situate field observations within broader institutional descriptions that are themselves treated critically in the analysis.

4.3 Ontology construction

The ontology was developed specifically for the Andean ritual domain rather than derived directly from existing standards. Partial alignment with the CIDOC Conceptual Reference Model was adopted to support interoperability while avoiding the imposition of conceptual structures incompatible with the granularity of field-based evidence [26].

The model defines core entity classes including *Festividad*, *EventoRitual*, *Lugar*, *LugarSagrado*, *Apu*, *Nacion*, *Danzante*, *Ukumari*, *Danza*, *Vestimenta*, *Instrumento*, *ObjetoRitual*, and *RecursoMultimedia*. Relations are organized across temporal, spatial, participatory, material, and provenance

dimensions. Temporal sequencing is encoded through *antesDe* and *sucedeInmediatamenteA*, spatial containment through *estaEn* and *contiene*, participation through *realizadoPor* and *participaEn*, and provenance through *documentaEvento*, *documentaLugar*, and *documentaParticipante*.

Temporal organization follows a dual structure derived from field observations: a sequence of named temporal frames, corresponding to five days and two nocturnal periods, and an ordering property, *tieneOrdenEvento*, that captures the sequence of events within each frame. This distinction allows the model to represent both the calendrical organization of the pilgrimage and the internal ordering of ritual actions.

A central modeling decision concerns the representation of dual event-place entities. Five entities in the graph are modeled simultaneously as instances of *EventoRitual* and *Lugar*: Lomada, Inti Alabado, Pachakunapata, Escalerachayoq, and the vehicular journey from Paucartambo to Mahuayani. This decision emerged from field-based observation and was refined through collaborative interpretation with a co-author who had previously participated in the ukukus. In these cases, events and places are co-constitutive: the event is defined through its spatial trajectory or site, and the place is constituted through the performance of the event.

From a formal perspective, this modeling decision also improved the internal consistency of the graph. The ontology was implemented and iteratively refined in Protégé [19], and logical coherence was evaluated using the *HermiT* reasoner. The introduction of dual event-place entities reduced inconsistencies in the representation of composite ritual processes, particularly those involving extended trajectories and named waypoints.

Spatial hierarchy is represented through transitive *estaEn* relationships, enabling multi-scale containment from local waypoints to broader structures such as sacred zones, glacier areas, and the Ausangate apu. The ontology presented here corresponds to an initial version focused on spatial, temporal, and participatory aspects of the pilgrimage, modeled primarily from the perspective of the ukukus. Other dimensions of the ritual system, including additional dances, naciones, and inter-festival relationships, remain only partially represented.

4.4 Provenance and versioning

Every assertion in the knowledge graph is associated with provenance metadata, including evidence type, acquisition date, fieldwork context, and responsible annotator. Evidence is categorized into three types: first-person observational data collected during the June 2025 fieldwork, community testimony obtained through structured interviews with ritual participants and authorities, and contextual documentary references, including UNESCO and Ministerio de Cultura materials.

This provenance structure allows each assertion to be traced back to its source and evaluated in relation to its evidentiary context. Within the retrieval framework, provenance metadata also makes it possible to distinguish between directly observed data, participant-reported knowledge, and externally documented descriptions, thereby supporting interpretability in the analysis of retrieved subgraphs [23].

The knowledge graph described in this study constitutes version zero of a planned iterative se-

quence. It provides a baseline representation constructed from the June 2025 Qoyllur Rit'i fieldwork and the contextual documentary references available at the time of writing. Subsequent versions are planned to incorporate additional field campaigns at Qoyllur Rit'i and the Virgen del Carmen festival in Paucartambo, enabling comparison across related Andean ritual systems within a shared ontological framework. This versioning strategy is intended to support empirical evaluation of how graph expansion and densification affect retrieval performance [9].

4.5 Ethical considerations

This study involves field-based research conducted in the context of the Qoyllur Rit'i pilgrimage in June 2025, complemented by prior engagement with the Virgen del Carmen festival in Paucartambo. As such, it includes direct observation, participation in ritual activities, and the collection of community-based knowledge.

All data collection followed informed consent protocols. Participants were informed of the academic purpose of the research, the non-commercial use of the data, and their right to withdraw at any time. Personal identifiers were recorded only with explicit permission, and sensitive cultural information was treated with particular care in accordance with good practices in cultural heritage research.

Competing interests

The author(s) declare no competing interests.

Data availability

The knowledge graph generated and analyzed during this study is available in the GitHub repository at https://github.com/javier-vz/graphrag_qoyllurity. The repository includes the RDF/Turtle graph files, as well as the entity and index representations used for retrieval.

Code availability

The code for the Graph-RAG pipeline and analysis scripts is available at https://github.com/javier-vz/graphrag_qoyllurity. The repository includes configuration files, query processing scripts, and evaluation components required to reproduce the results presented in this study.

Informed consent statement

This study involved participant observation and interviews conducted with informed consent. Participants were informed of the academic purpose of the research and the use of the data, and participation was voluntary. No sensitive personal data were collected.

References

[1] R. Alturki et al. Knowledge graph applications in cultural heritage: A ROSES-based systematic review. *Information*, 17(3):269, 2025. doi: 10.3390/info17030269.

[2] Payal Arora. Creative data justice: a decolonial and indigenous framework to assess creativity and artificial intelligence. *Information, Communication & Society*, 27(11):2231–2247, 2024. doi: 10.1080/1369118X.2024.2420041.

[3] Gabriela Arriagada-Bruneau, Claudia López, and Mauricio Mendoza. *Ethics in Artificial Intelligence and Information Technologies*. CRC Press / Taylor & Francis, 2025. ISBN 9781032594187.

[4] Valentina Carriero, Aldo Gangemi, Michele Mancinelli, Luca Marinucci, Andrea Giovanni Nuzzolese, Valentina Presutti, and Chiara Veninata. Arco: The italian cultural heritage knowledge graph. In *The Semantic Web – ISWC 2019*, pages 36–52. Springer, 2019. doi: 10.1007/978-3-030-30796-7_3.

[5] Edward Davis and Bahareh Heravi. Linked data and cultural heritage: A systematic review of participation, collaboration, and motivation. *Journal on Computing and Cultural Heritage*, 14(2), 2021. doi: 10.1145/3429458.

[6] Catherine D'Ignazio and Lauren F. Klein. *Data Feminism*. MIT Press, 2020.

[7] Siyu Duan, Jun Wang, Hao Yang, and Qi Su. Disentangling the cultural evolution of ancient china: a digital humanities perspective. *Humanities and Social Sciences Communications*, 10: 310, 2023. doi: 10.1057/s41599-023-01811-x.

[8] Daniel Edge and et al. From local to global: A graph rag approach to query-focused summarization. *arXiv preprint*, 2024. URL https://arxiv.org/abs/2404.16130.

[9] T. Fan, H. Wang, and T. Hodel. Cichmkg: A large-scale and comprehensive chinese intangible cultural heritage multimodal knowledge graph. *Heritage Science*, 11:115, 2023. doi: 10.1186/s40494-023-00927-2.

[10] Achille Felicetti, Ahmed Himmiche, and Marco Somenzi. Knowledge graphs and artificial intelligence for the implementation of cognitive heritage digital twins. *Applied Sciences*, 15 (18):10061, 2025. doi: 10.3390/app151810061.

[11] Anna Foka, Gabriele Griffin, Diego Ortiz Pablo, et al. Tracing the bias loop: Ai, cultural heritage and bias-mitigating in practice. *AI & Society*, 40:5835–5847, 2025. doi: 10.1007/s00146-025-02349-z.

[12] Tianyu Gao, Xingcheng Yao, and Danqi Chen. Simcse: Simple contrastive learning of sentence embeddings. In *EMNLP*, 2021. URL https://arxiv.org/abs/2104.08821.

[13] Tarleton Gillespie, Pablo J. Boczkowski, and Kirsten A. Foot. *Media Technologies: Essays on Communication, Materiality, and Society*. The MIT Press, 01 2014. ISBN 9780262319461. doi: 10.7551/mitpress/9780262525374.001.0001. URL https://doi.org/10.7551/mitpress/9780262525374.001.0001.

[14] Yuexin Huang, Suihuai Yu, Jianjie Chu, Hao Fan, and Bin Du. Using knowledge graphs and deep learning algorithms to enhance digital cultural heritage management. *Heritage Science*, 11(1):204, 2023. doi: 10.1186/s40494-023-01042-y.

[15] Vladimir Karpukhin, Barlas Oguz, Sewon Min, and et al. Dense passage retrieval for open-domain question answering. In *EMNLP*, 2020. URL https://arxiv.org/abs/2004.04906.

[16] Angelie Kraft and Edwin Soulier. Knowledge-enhanced language models are not bias-proof: Situated knowledge and epistemic injustice in AI. In *Proceedings of the 2024 ACM Conference on Fairness, Accountability, and Transparency (FAccT '24)*, pages 1433–1445, 2024. doi: 10.1145/3630106.3658981.

[17] Shiyun Long and Wei Li. A chinese named entity recognition dataset for intangible cultural heritage. *Scientific Data*, 13:335, 2026. doi: 10.1038/s41597-026-06700-x.

[18] Zoila S. Mendoza. *Qoyllur Rit'i: Crónica de una peregrinación cusqueña*. La Siniestra Ensayos, 2021.

[19] Mark A. Musen. The protégé project: a look back and a look forward. *AI Matters*, 1(4):4–12, 2015. doi: 10.1145/2757001.2757003. URL https://doi.org/10.1145/2757001.2757003.

[20] Boci Peng, Yun Zhu, Yongchao Liu, Xiaohe Bo, Haizhou Shi, Chuntao Hong, Yan Zhang, and Siliang Tang. Graph retrieval-augmented generation: A survey. *ACM Transactions on Information Systems*, 2024. doi: 10.1145/3777378.

[21] Nils Reimers and Iryna Gurevych. Sentence-bert: Sentence embeddings using siamese bert-networks. In *EMNLP*, 2019. URL https://arxiv.org/abs/1908.10084.

[22] Guillermo Salas. Climate change, moral meteorology, and local measures at quyllurit'i, a high andean shrine. In *Understanding Climate Change through Religious Lifeworlds*, pages 44–76. Indiana University Press, 2021.

[23] Silvia Tiribelli, Stefano Pansoni, Emanuele Frontoni, and Benedetta Giovanola. Ethics of artificial intelligence for cultural heritage: Opportunities and challenges. *IEEE Transactions on Technology and Society*, 5(3):293–305, 2024. doi: 10.1109/TTS.2024.3432407.

[24] UNESCO. Pilgrimage to the sanctuary of the lord of qoyllurit'i, 2011. URL https://ich.unesco.org/en/RL/pilgrimage-to-the-sanctuary-of-the-lord-of-qoyllurit-i-00567.

[25] Javier Vera-Zúñiga, Felipe Urbina-Parada, and Dina Cornejo-Meza. Mapping semantic bias in unesco intangible heritage metadata through community detection in south america. *npj Heritage Science*, 14:133, 2026. doi: 10.1038/s40494-026-02400-2.

[26] Tong Wei and Yuqi Chen. A methodology for building domain ontology of cultural heritage. *Digital Scholarship in the Humanities*, 38(4):1710–1719, 2023. doi: 10.1093/llc/fqad045.

[27] Liang Xu, Lu Lu, and Minglu Liu. Construction and application of a knowledge graph-based question answering system for nanjing yunjin digital resources. *Heritage Science*, 11:222, 2023. doi: 10.1186/s40494-023-01068-2.

[28] Hua Yuan, Yuhan Li, Baohui Wang, Kai Liu, and Jun Zhang. Knowledge graph-based intelligent question answering system for ancient chinese costume heritage. *npj Heritage Science*, 13(1):198, 2025. doi: 10.1038/s40494-025-01776-x.

[29] W. Zhang et al. CIDOC CRM-based knowledge graph construction for cultural heritage using large language models. *Applied Sciences*, 15(22):12063, 2025. doi: 10.3390/app152212063.